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Metaphysic AI: Rethinking the Value of Human Expertise

The most important thing I think about is – what will the internet look like in ten years? There will be a hyper-realistic content layer for all the experiences we currently prefer to have in person. This content will be created from real-world data – video, audio, voice, the data from our private spaces and our perception of reality – all streaming into foundational AI content creation models to generate hyper-realistic personalized experiences. This personal data should fundamentally belong to, and be owned by, each and each individual. How do we ensure individuals are at the center of controlling and consenting to that process?

– Tom Graham, CEO and Co-Founder, Metaphysic¹

Introduction

In January 2025, Thomas Graham, co-founder of Metaphysic, a leading AI generative video company, sat in his London office contemplating a range of issues about the future of digital identity and content creation. His company had recently achieved a significant milestone with the movie 'Here' – the first major studio production to use AI throughout an entire feature film, and starring a de-aged performance from Tom Hanks as his younger self circa 'Big' (1988). Working with the cast and crew to unearth new creative storytelling possibilities on the fly, Metaphysic used a variety of in-house AI products. These included tools that allowed actors to be de-aged live and in real time on set, as well as latent-space facial performance enhancement.

But, Graham wasn't just thinking about Hollywood. He was grappling with a more fundamental question: in a world where a person's likeness (their face, their voice) could be perfectly recreated with AI, who should control the rights to that digital identity? This struck at the heart of what Hollywood had long recognized through "separated rights" – a concept established in the 1950s to ensure that creators maintain ownership of their work beyond specific contracted uses. In the context of AI, these rights determined who controlled and profited from an actor's digital recreation, potentially for decades after the original performance.

Founded in 2021, Metaphysic had developed technology that could convert what was once inseparable from a performer's physical presence – their appearance, expressions, and movements –

Professors Zoe B. Cullen and Shikhar Ghosh and Research Associate Shweta Bagai prepared this case. It was reviewed and approved before publication by a company designate. Funding for the development of this case was provided by Harvard Business School and not by the company. HBS cases are developed solely as the basis for class discussion. Cases are not intended to serve as endorsements, sources of primary data, or illustrations of effective or ineffective management.

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into digital code capable of creating new performances. The company first gained public attention through a series of viral TikTok videos (@deepmcrruise) featuring an AI version of Tom Cruise, which garnered over 420 million views on TikTok alone and over 2 billion across all platforms. Building upon this early viral success, Metaphysic further showcased their technology on America's Got Talent (AGT) in 2022, where their photo-realistic Elvis Presley performance helped them finish in 4th place, and generated another 115 million views across social platforms. The company built its business providing high-end content creation products for enterprise (including all major Hollywood studios) along with developing a platform for individual rights management, underpinned by a strong ethical stance on data protection.

Metaphysic operated in an environment where traditional legal frameworks struggled to keep pace with technological capabilities, particularly regarding AI-generated content that was becoming increasingly indistinguishable from reality. In 2023, the SAG-AFTRA strike comprising over 160,000 actors and performers,² led to studios agreeing to unprecedented protections around AI and performance data, requiring explicit informed consent and fair compensation for using a performer's digital likeness. These protections built upon Hollywood's established negotiation frameworks, which had evolved over decades to protect creative rights in step with changes in technology. As a leading advocate for the rights of performers to control their AI-generated performances, this framework allowed Metaphysic to work with both artists and studios. On one hand, Metaphysic helped actors enhance their performances with AI, and on the other, they developed the core technologies used by studios. Protections however did not extend to the broader entertainment industry or to other professions, where work products often fully belonged to employers.

Graham knew that the implications of their technology extended far beyond Hollywood. This ability to digitally capture and recreate human likeness raised profound questions about control, consent, and ownership of expertise in the digital age. In the absence of a protective structure of industry-wide collective bargaining, how would rights and compensation be negotiated? How could Metaphysic build a scalable business model that respected individual rights while delivering value to clients?

Founding Story

Origin

In 2018, deepfakes—digital creations that almost exactly replicated a real person—were hardly in the vocabulary of the general public. The technology was primarily associated with harmful applications, particularly face-swapped pornographic content, often targeting celebrities. In the political context, BuzzFeed created a viral announcement featuring a deepfake of President Barack Obama seemingly making outrageous statements, intended to warn about the technology's dangers.³ The term “deepfake” combined “deep learning” and “fake,” implying content that was artificially generated and had never existed in reality. This perception was shaped by its growing association with misinformation and disinformation campaigns.

The increasing prevalence of the word reflected humans' ability to extract the tacit expertise and likeness of a person. On social media, deepfakes tended to target celebrities and influencers, with artists like Justin Bieber having their voice used without consent to create provocative songs about Sean “Diddy” Combs' parties, that surfaced in 2024.⁴ Two categories of deepfakes emerged: those that created entirely fabricated content (like superimposing someone's face onto another person's body), and those that recreated a person's full likeness to perform actions they hadn't actually done, but could theoretically do (like an actor in a new scene).

Drawing on his academic background in internet law and society and a decade of entrepreneurial experience in big data and AI, Graham co-founded Metaphysic in 2021 to explore the potential of deepfakes as a legitimate tool for the entertainment industry. Graham had previously worked on how emerging technologies changed social and economic institutions, and had been focused on the importance of safeguarding personal data, especially the kind that could be used in AI training. He elaborated, “The origin of our company was trying to think about what to do with that as a status quo.”^a The company focused on ‘ethical AI’ “long before the talk of generative AI,”⁵ distinguishing itself from later companies that simply built applications on top of foundation AI models.

The company’s potential soon became clear, when a series of videos with an AI version of Tom Cruise playing golf, telling jokes, and performing magic tricks went viral on TikTok. The videos, posted under the account @deeptomcruise were among the first truly photo-realistic AI-generated videos and were created through a collaboration between a visual effects artist, Christopher Ume, his brother Kevin, and Miles Fisher (who provided the underlying Tom Cruise impersonation). (See **Exhibit 1: Tom Cruise deepfake**). When Graham saw the first @deeptomcruise video on the day it was originally posted, he noticed that it had tens of millions of views from people who genuinely believed they were watching the real Tom Cruise. Graham immediately reached out to the Ume brothers to understand why their deepfake was so much better than anything seen to date. Within weeks, Metaphysic was incorporated with the mission to build the software and models to scale photo-realistic AI content to every person on Earth. The founding team comprised Graham, the Ume brothers, and Fisher along with serial tech entrepreneur, Martin Adam, and VFX (visual effects) industry veterans, Jo Plaete and Ed Ulbrich.

The technical sophistication behind these videos highlighted why few companies operated in this space. The creation of these videos was far from simple. Christopher Ume explained, “You can’t do it by just pressing a button.”⁶ Each clip required weeks of work, combining open-source algorithms with traditional CGI (computer-generated imagery) and VFX tools. The process of training AI models would take weeks using a NVIDIA RTX 8000 GPU, followed by days of frame-by-frame adjustments to achieve a lifelike effect. “The most difficult thing is making it look alive,” Ume noted. “You can see it in the eyes when it’s not right.”⁷ The challenge of creating realistic human faces, long considered one of the most difficult aspects of CGI, represented just the beginning of this transformation, setting the standard for Metaphysic’s approach to AI-generated content.

Early Growth and Funding

Metaphysic’s early work extended beyond the Tom Cruise videos. The company collaborated with brands and organizations, including Gillette, 60 Minutes and The Belgian Football Association, along with artists including Drake and Aloe Blacc. Their focus was not just on entertainment, but on building what Graham described as “connections among people, and creating a more seamless interface between reality and the time we spend online.” He explained, “How do we bring regular people into content? How do we create content that’s meaningful, interesting, and relevant? How do we make it look realistic, rather than like a meme or cartoon? That’s what we focus on—the research, the technology, and development.”⁸

In January 2022, the company secured a \$7.5 million seed investment and, later in 2022, a further \$22 million Series A investment (via SAFE notes) from investors including Section 32, 8VC, Liberty Global, Rakuten Capital, TO Ventures, Winklevoss Capital and a number of prominent figures from

^a Unless noted otherwise, quotes from Tom Graham are sourced from an interview conducted by the casewriters in September 2024.

across the sports, entertainment and music industries to expand their work on AI content creation tools.⁹ Starting with a team of 15 to 20 people, Metaphysic positioned itself at the intersection of entertainment and ethics, emphasizing the importance of informed consent and responsible development of generative AI technologies. Graham believed that there was no public utility or fair use rationale for taking someone's exact likeness and using it without permission. "It just doesn't feel right that a company should own the ability to recreate a version of your child's face and voice. On every level, it doesn't feel right that a government should have the ability to do that either." He had previously advocated for copyright law reform to recognize individuals' rights to their AI-generated likeness. Moreover, Graham supported the notion that individuals should own and control how their AI avatars and voices were used, with legal mechanisms to enforce these property rights, and challenge unauthorized deepfakes when they appeared.

Technology and Implementation

The Hollywood Breakthrough

By 2022, Metaphysic had proven their technology's capabilities through high-profile demonstrations, most notably their America's Got Talent (AGT) appearance featuring a live, AI-generated performance by Elvis Presley. (See **Exhibit 2: Elvis Presley with the AGT Judges**). Their proprietary technology generated high-fidelity, photo-realistic face swaps in real-time. Graham elaborated, "The aim of our performance was always to harness the power of AI and synthetic media to create entertaining and meaningful content to delight the audience."¹⁰ One of the judges, Simon Cowell, stressed the surreal nature of watching himself perform something he did not, or potentially could not, do. He reacted with, "I've always said, is there such a thing as a perfect contestant? We've just found them!"¹¹

The entertainment industry soon found itself at the epicenter of a broader technological revolution. The emergence of large language models (LLMs) and generative AI demonstrated that human expertise – whether in writing, coding, or acting – could be captured, processed, and replicated at scale. This transformation was particularly visible in Hollywood, where advances in synthetic media technology had moved from curiosity to commercial reality. As studios grappled with the creative and economic implications of AI technology, Metaphysic's capabilities caught their attention. A pivotal moment emerged in late 2022 when Metaphysic became the first AI company to partner with Creative Artists Agency (CAA). This partnership signaled a willingness for Hollywood to experiment with AI technology.

But, Metaphysic needed to prove that their technology could work at Hollywood scale. The opportunity came when Robert Zemeckis, the Oscar-winning director of 'Forrest Gump' approached them about his film 'Here.' The project presented a unique challenge. The film, based on Richard McGuire's graphic novel, required Tom Hanks and Robin Wright to appear at various ages, from 18 to their 80s. Traditional approaches were impractical: conventional makeup could only take a decade off 68-year-old Hanks, and standard CGI de-aging would require hundreds of VFX artists and tens of millions of dollars – a cost that had previously rendered the project unviable.

The Production System underlining 'Here'

Metaphysic developed a groundbreaking on-set system that transformed traditional filmmaking workflows. The system worked by creating high-resolution AI models of Hanks, Wright and two other actors across a total of eleven age/actor characters. Each model was trained on licensed historical data from when Hanks and others were younger as well as proprietary facial scans captured by Metaphysic.

In combination, these relatively modest datasets allowed AI models to encode an actor's performance, their facial expressions, movements, and vocal patterns. On set, these models were also constantly fine-tuned on all available footage shot while filming — allowing for the models to better understand context and lighting data. Metaphysic's algorithms allowed for the creation of digital models that could generate new, photo-realistic performances of Hanks and others at any age. Graham described, "We input young 20-year-old Tom Hanks on top of Tom Hanks' performance which is a large technical feat. We used Tom Hanks' real performance to provide the underlying destination data for how things are animated, how the face moves, to create and capture that naturalism of putting a synthetic version on top."¹² (See **Exhibit 3: Characters in 'Here'**).

Not only did Metaphysic use AI to generate the final high-resolution shots in post-production, but they also deployed live, real-time versions of their technology while actually shooting on location. This production system allowed Zemeckis and his crew to monitor two different feeds simultaneously: one showing the raw camera footage of the actors as they appeared in reality, and another filtered through their AI tools showing the actors at whatever age the scene required. This real-time feedback transformed the production process. "It was crucial that the cast could see it," Zemeckis noted, "because then they could adjust their performance. They could say, 'Oh, I see, I've got to make sure I'm moving like I was when I was 17 years old.'"¹³ They didn't have to imagine it. Ed Ulbrich, Metaphysic's chief content officer, observed, "I could take my entire team from my former places of employment, I could put them on for eternity using the best CGI tools money can buy, and you can't deliver what we're showing you here. And it's happening in milliseconds."¹⁴

Rather than replacing existing production methods, Metaphysic's technology complemented traditional filmmaking craft. The actors performed in full period costume, while the makeup department made physical adjustments — taping back loose skin, adjusting ears — to ensure the AI-generated young faces would match the actors' real heads.¹⁵ This hybrid approach maintained the authenticity of performances while enabling previously impossible visual transformations. Zemeckis claimed that after trying several technologies to make the de-aging process possible, Metaphysic were the global leaders in feature-quality AI content.¹⁶ He added that, without Metaphysic, the economics didn't work for a character-driven story without "capoes or explosions or aliens or superheroes or creatures."¹⁷ The film was completed for about \$50 million — less than a quarter of typical Marvel movie budgets.¹⁸ Zemeckis described, "I've always been attracted to technology that helps me to tell a story. With 'Here,' the film simply wouldn't work without our actors seamlessly transforming into younger versions of themselves. Metaphysic's AI tools do exactly that, in ways that were previously impossible!"¹⁹

Business Model

Since the outset, Metaphysic's mission had been to build the software and technology to support automated AI-generated content. However from 2021, few customers were willing to pay for relatively expensive, high production-quality content. While most companies experimenting with AI-generated content focused on consumer platforms serving low-quality user-generated AI images, video and memes — this was not Metaphysic's strength. Instead, they naturally gravitated towards the relatively small VFX market in Hollywood. Metaphysic established itself as the leader in production-quality AI content for major films. Graham explained, "We focus on creating photo-realistic AI-generated content — content that looks exactly like the real world, as if it were shot with a video camera. That's incredibly difficult because we're producing those pixels with algorithms."²⁰ He elaborated, "No one else does any kind of volume [of shots] in this space of creating such content, because it's really difficult. You need a lot of software to do it at scale or in a cost-effective way that...is also good enough to show

at something premier like a Hollywood film.” While AI models, including multi-modal LLM and diffusion-based models could generate images and video, they lacked the control and consistency required for professional productions. Graham described, “We’ve been able to use visual effects in movies for a long time, but they never quite look truly real. There’s always some level of the uncanny valley. Today, with generative AI—especially the technology we build—you can create a version of reality that never happened, but which people perceive as real because it is so immersive.”²¹ Metaphysic structured its business around two distinct revenue streams, each with a unique approach to data ownership and rights management.

Metaphysic Pro – A Data Platform

The first revenue stream was a B2B2C service where individual clients (such as actors or performers or anyone with valuable IP) could store and control their likeness data, the AI avatars, and personalized models created with that data. Ultimately, the best sales channel to onboard individual users was to partner with performers’ unions, players’ associations, agencies, sports and management groups. This service functioned like a specialized Dropbox for identity data, allowing users to retain complete ownership of their digital assets, which could be used in the future for any purpose. Graham explained: “You’re not trying to productize somebody and sell it back to them. You’re trying to give them a way to safeguard the means of production of their future AI performance.” Beyond helping people or IP holders build a valuable portfolio of data assets—whether for personal protection or monetization—the platform also protected AI likenesses through legal frameworks like copyright. Metaphysic developed a process to help people create their AI likeness and voice model, and submit them for copyright registration.

As of early 2025, the U.S. Copyright Office had not recognized the AI generated elements of a digital avatar as eligible for copyright protection. However, the office was proactive in evolving copyright regulation in a direction that Graham hoped would make this possible.²² He described:

If someone creates an unauthorized deepfake of you and posts it online, you need a way to take it down. Copyright law offers one of the fastest takedown methods, but only if you’ve registered your likeness as copyrighted material. That’s why I personally went through the process to register my own AI likeness with the U.S. Copyright Office. [...] It’s a form of legal activism. We’re pushing the boundaries to establish legal rights for individuals over their AI likeness.²³

The company’s approach was distinctive: they maximized the potential of performance data to create the absolute best deepfake on the market and recognized that today’s performances needed to be captured and stored, with a high level to detail, to be used effectively in training AI models in the future. Metaphysic then let their client, the artist, license their deepfake creations to a wide variety of outlets including studios, brands and anyone that wished to use their AI likeness within a consent-based and commercial framework. At the same time, they could set themselves up to challenge unauthorized deepfakes by both applying for copyright registration or claiming the underlying copyright in their AI likeness by virtue of the act of having created it in the first place.

It is notable that, even without asserting any specific legal basis for copyright protection, people were often successful in issuing takedowns against platforms hosting unauthorized deepfakes by the mere assertion of ownership over or appropriation of their likeness. Proactively creating a person’s AI likeness, owning the models, and training data became the fundamental building blocks for the protection of creators from unauthorized generative AI and identity appropriation. In this way, Metaphysic established the foundations for an expanded right to publicity that catered to the realities of new AI-driven content creation by allowing people to create the underlying property, including

datasets, models and AI avatars, the ownership of which was the basis for copyright and other infringement claims and remedies.

Professional Services for Studios

Metaphysic's second revenue stream was from providing visual effects services (VFX) directly to studios, creating AI-generated content for major productions. In these B2B relationships, Metaphysic developed a novel approach to intellectual property and data ownership. “Normally what happens in a VFX contract is that they own everything: the model, all the data and photography, photogrammetry, used to create the 3D model, everything,” explained Graham. However, Metaphysic carved out a unique position: while studios owned both the source data and the final content output, Metaphysic retained ownership of their AI models. This arrangement protected Metaphysic's proprietary technology while allowing studios to fully own and utilize the finished content. Examples included Metaphysic's work on ‘Here,’ ‘Alien: Romulus’ and ‘Furiosa: A Mad Max Saga.’

The company extended its technology beyond visual effects into voice synthesis for the music industry. Their capabilities were publicly demonstrated at the Video Music Awards (VMA) in two ways: first, they won the Visual Effects VMA for their work creating young Slim Shady in Eminem's music video “Houdini” and other supporting content; and second, during Eminem's opening performance on stage, they created a world-first live broadcast of a real-time AI hologram of Slim Shady performing alongside Eminem. Metaphysic also worked with a large number of musicians to create high-quality audio and voice models for professional music production. Graham described, “We're not just creating entertainment content for movies, commercials, or AI-generated concerts for music artists. Our goal is to build technology that scales so that everyone on Earth can use it. Ten years from now, or even sooner, I believe more than 90% of all the content people consume on their phones will be AI-generated.”²⁴

Hollywood: The Original Codified Expertise Economy

Hollywood represented an industry with a unique, long-standing history of codified labor — where creative expertise had been captured in tangible, reproducible assets for nearly a century. While most industries relied on embodied knowledge that remained with the worker, Hollywood naturally encoded creative contributions into concrete forms: screenplays, footage, and recordings. A writer's expertise became fully embodied in their script, while an actor's craft was preserved frame by frame in film. This fundamental characteristic — the inherent codification of creative work — drove the early development of sophisticated rights frameworks (and would later serve as models for an AI-driven future).

History of Rights over Products of Codified Expertise

Recognizing this reality, creative expertise sought to carve out rights over their “disembodied” work products. First, Hollywood Guilds^b worked to establish attribution for any codified labor that could not be easily mapped to its creator once disembodied (actors on screen were the exception). The Writers' Guild established a peer-review credit allocation system as early as 1939, whereby the Guild reviewed the version history of a script and determined partial or full credit to individual writers, assigning their official bylines. By 1953, writers had secured “separated rights” — a revolutionary

^b Hollywood Guilds included different entertainment industry labor unions such as Writers Guild of America (WGA) for writers; Screen Actors Guild-American Federation of Television and Radio Artists (SAG-AFTRA) for actors; Directors Guild of America (DGA) for directors, and others.

concept ensuring creators maintain ownership of their work beyond the specific contracted use. It meant that any work that was not explicitly described in the work-for-hire contract would be retained by the creator of the work, including rights over all future use cases.

This framework acknowledged a crucial truth: once creative labor was codified, it gained a permanence requiring special protection. When expertise was captured in scripts or performances, it could be reproduced and monetized indefinitely, in ways impossible to anticipate at creation. These separated rights ensured that when new technologies emerged—from television to streaming platforms—the original creators retained a stake in how their codified expertise was utilized.

Adaptive Labor Market Structure

Hollywood offered profound insight into how labor markets transformed when expertise became codified. Economic theories of the firm, developed by Bengt Holmstrom, Oliver Hart, and Robert Gibbons, explained that traditional employment existed primarily to resolve fundamental contracting problems: employers could not fully observe effort, outputs remained uncertain, and humans could not be owned or indentured. These limitations led to firms emerging as governance structures with ongoing relationships when simple market transactions proved insufficient.

However, where expertise was codified—transformed into observable, transferable assets like scripts, footage, or AI-ready data—these traditional frictions disappeared. Codified expertise could be precisely specified in contracts, objectively evaluated, and traded like other commodities. Consequently, Hollywood evolved not into a traditional employment model but into what was recognized as the original “gig economy.” Production companies assembled specialized teams for specific projects, with employment typically lasting weeks rather than years. Contributors brought their codified expertise to projects, negotiated compensation (including future rights), and moved on to their next opportunity.

Post-AI Collective Bargaining Agreement of 2023

During the early adoption phase of Large Language Models (LLMs), creative expertise Guilds and Studios met to re-negotiate their collective bargaining agreements. The lack of comprehensive data protection outside Hollywood created a stark contrast to the protections negotiated by entertainment unions, raising questions about how other industries might develop similar frameworks. (See **Exhibit 4: Status of Digital Data Protection**). Building on their history of negotiated rights, in 2023, the Writers' Guild kicked off the strike and negotiations, setting the stage for SAG-AFTRA (the actors/performers Guild, with over 160,000 member) and other Guilds to follow suit. The Writers' Guild of America West gave top billing to AI during contract renegotiations with Hollywood studios. After five months of striking, they reached an agreement with the Alliance of Motion Picture and Television Producers (AMPTP) to enshrine new rules for its contracts with employees. This featured groundbreaking provisions for the use of artificial intelligence in the writing process of films and shows and, guaranteed minimum employment (for the first time).²⁵ (See **Exhibit 5: Excerpts from Proposed Package to WGA**).

The guidelines stated that works created by AI were not considered literary material. The Writers' Guild demanded that credit and separated rights not be affected by generative AI. Thus, production companies could not undermine a writer's pay by providing an AI-generated script as a basis for the screenplay and claiming the writer was only rewriting the script (which would result in lower pay). The writer was still considered the first writer on the screenplay in such a scenario and the writer retained rights over their products in the same manner as before. The studio or production company established internal guidelines for using AI in the creative process. However, if a project was covered under the MBA (Minimum Basic Agreement), guidelines around defining attribution took precedence.

In addition to retaining creative rights, artistic creators also had to grapple with another reality: anecdotal evidence suggested that a writers' room could be up to 10 times more efficient when using AI tools to aid creation, mimicking the benefits that coders experienced. Thus, many fewer writers or hours of work might be necessary for the same creative output. Perhaps the largest departure from past negotiations, the Guild demanded that the studios respect minimum staffing levels for each project, and on August 11, 2023, the studios agreed to this.

Following the Writers' Guild agreement, on December 5, 2023, SAG-AFTRA completed an agreement with AMPTP which stipulated that artist consent was required for the use of the voice or likeness of individuals if the artist was not directly contracted to produce the work. With required consent, of course, came bargaining power and compensation.^{26c} These steps forward largely focused on the products of generative AI models. Around the same time as the Hollywood Guild negotiations, Congress began exploring legislation to protect worker data rights as well, though no comprehensive bill had successfully passed both chambers in 2024. (See **Exhibit 6: Upcoming Federal Legislation**).

Metaphysic: Negotiating Digital Performance Rights

Contributions and Rights of Multiple Parties

For actors and performers, visual and audio data captured much of their valuable labor. This was in contrast to consultants, venture capitalists or engineers, whose valuable expertise might be derived from the data on their computers or their monitored interactions and series of judgements. The economic value of a deepfake rested on its ability to codify a person's expertise, whatever it may be. In this context, Metaphysic helped professionals codify their expertise to the highest quality, and in select use cases, could enable people to be proactive in designing, protecting and selling their codified expertise. The transformation of work into data presented novel negotiation challenges. Unlike traditional work products, data-derived performances blurred ownership boundaries. (See **Exhibit 7: When Work Becomes Data**).

While studios had long maintained archives of actor performances, these assets gained new significance. What was once simply footage for potential reuse had become training data that could power AI systems capable of generating new content. This shift fundamentally changed the view of an actor's work and its long-term value. Hollywood had a leg up in determining rights around unanticipated future use cases via "separated rights." Indeed, Gracie Studios revived the very footage being used to create 'Here' with the advent of the internet and streaming capabilities, striking new licensing agreements for usage of 'Big.' The very same clauses that determined rights for streaming would lay the groundwork for parties to contract over AI uses.

Performance Data Value and Classification

Metaphysic, the new player in town, established its contribution and ownership rights. The advances in AI were exemplified by their work on 'Here' which demonstrated the complexity of this new data landscape. This wasn't merely about de-aging — it represented a fundamental shift in how performance data could be captured, processed, and redeployed. Metaphysic developed a sophisticated understanding of data value, distinguishing between traditional tracking data and 'core identity' (or likeness) data which was considered to be higher value due to its ability to generate new

^c In addition to the agreement that the organization reached with AMPTP, SAG-AFTRA is actively lobbying in D.C. to create federal property rights for individuals' image and likeness through the NO FAKES Act.

content. Graham explained: “If we are talking about enough video of your face or audio of your voice to feed into a system that generates new content indistinguishable from reality — that’s substantially different from mere behavioral data, like that derived from cookie tracking over a discrete period of time.”

Metaphysic’s system required multiple categories of data, starting with the foundational elements: high-resolution video captures of Tom Hanks from multiple angles, voice recordings spanning different emotional ranges, and detailed motion capture of facial expressions. Licensed historical footage spanning Hanks’ four-decade career was also needed. Each period represented a different ‘version’ of the actor that could be synthesized for new scenes. However, the value of this data extended beyond raw footage. Metaphysic’s AI models were nuanced enough to also capture ‘performance intelligence’ — the subtle patterns that made a Tom Hanks performance distinctively his own. Elements such as his characteristic head tilt when delivering emotional dialogue, the timing of his smile, the rhythm of his speech needed to be understood and codified by the AI algorithms. This deeper layer of data represented the essence of the actor’s craft. Graham added, “It’s not just about the algorithm—it’s about the dataset. The actor should own and control the data used to generate their likeness. Sometimes, we need highly controlled studio scans to build a refined dataset, ensuring the final AI-generated version looks indistinguishable from reality.”²⁷

The monetary implications of this transformation challenged industry frameworks. Traditional media contracts had long recognized different types of value capture. An actor’s performance in a theatrical release earned different residuals than streaming, international distribution, or merchandising. However, AI-generated performances created using historical data presented new challenges for valuation. If an AI system trained on data from a 1985 performance created a new scene in 2024, what was the appropriate compensation model? Was this equivalent to reusing old footage, or was it creating something entirely new? (See **Exhibit 8: Data Value and Attribution**). AI systems didn’t simply reuse discrete segments; they learned patterns from entire performances to generate something new. This fundamental difference made existing compensation models inadequate, forcing the industry to reconsider how it valued and compensated for the use of performance data. Emerging compensation models such as micropayments and blockchain verification presented potential negotiation frameworks that could address the unique challenges of AI-generated content. (See **Exhibit 9: New Compensation Models**).

Structuring the Tom Hanks Contract

The complexity of creating young Tom Hanks required careful navigation of rights and permissions. Metaphysic operated through the studio’s contract with Tom Hanks, rather than contracting directly with the actor. The process required the studio to carefully navigate historical content rights. In a project involving de-ageing, a studio or production company would need to license imagery from historical movies featuring a younger performer from the owner of those movies. Metaphysic would then use this licensed data to train their AI models. The company established clear boundaries around data and model ownership. While a studio would own both the source data (licensed from previous films) and the final output, Metaphysic retained ownership of the AI models but not the right to recreate any actor’s performance in the future without permission from all parties involved. As Graham noted, “The [AI] model is inextricable from our core technology.”

This arrangement implied that if another studio or client wanted to use Hanks’ likeness for a different project, Metaphysic would train a new model from scratch using newly licensed data. It was Metaphysic’s policy to collect consent from all the parties involved in creating the original content. For example, Tom Hanks, as well as the studio which created the original footage of young Tom Hanks

being used, would both need to consent to the terms for creating an AI version of Hanks, de-aged to look like his character in 'Big' (1988). Those terms would fully specify the new content, the owners of that new content, and the distribution of revenue.

Separately, via their Metaphysic Pro platform, they were able to establish a different relationship with performers, like Hanks, giving them direct control over data that they collected themselves and owned outright. Within the Pro platform, users could upload data of themselves and create an AI generated avatar, including an AI model trained on their data and rendered video content. Regarding usage, Metaphysic stated that users could not use their avatar content off-platform except for social media posts. They warned users that third-parties may scrape and steal the likeness data from these social media posts. However, Metaphysic assured that no third party would ever have access to the original models or datasets stored on the Pro platform itself. A Pro contract could allow Metaphysic to legally pursue anyone who copied AI content and deepfakes created using the platform. If Metaphysic was not contracted to enforce the individuals rights, then it was up to the artist to pursue this under their union contract or potentially through U.S. copyright law. At this time, they may have to rely on the Lanham Act, torts for invasion of rights of publicity and privacy.

In standard labor markets, the employer likely held exclusive rights to the product used to train the AI model. While Tom Hanks may garner unusual protections on account of his public persona and broad fan base, most employees transfer all products of their labor to their employer for future use cases. This traditional model faced new challenges with AI-generated content. Even with licensed footage from 'Big,' studios couldn't automatically create deepfakes without potential legal challenges. This created an opportunity for forward-thinking performers to take control by collecting their own data and create an equally compelling deepfake. Had Tom Hanks proactively captured and retained high-quality data of his performances in his 20s, he could have directly licensed this superior dataset to Metaphysic and studios, bypassing the need to use footage owned by former employers. This disintermediation would give performers significant economic leverage—self-collected data would likely produce better quality AI recreations at lower costs, as performers could continuously capture optimal reference material. A Pro contract with Metaphysic would protect such employee-generated self-replicas, in advance of any other.

Beyond Hollywood: Future of AI-Generated Content

Metaphysic stood at the forefront of a new era in digital identity and content creation. The challenges they faced in managing performance rights and data ownership would inevitably trickle down to other companies and industries. As he contemplated Metaphysic's future beyond Hollywood, fundamental questions emerged about data rights in an AI-driven world.

The company had built its business model on the foundation of Hollywood's "separated rights" framework—established through decades of collective bargaining that required creator consent and compensation. These collective action agreements prevented the employers (i.e., studios) from exploiting the data created on-the-job in perpetuity, in whatever way they choose, fully circumventing the consent of individual contributors. Precisely because studios did not by default have full control over reusing the footage generated in the course of producing a particular film, Metaphysic could intervene in the licensing of those data and the models produced by those data. Metaphysic prioritized the artist in their business model, considering the protection of the artist's work their primary duty. However, as AI technology democratized and spread to other sectors, this protective framework would not exist. In most industries, employers already owned workplace data by default, and extensive digital monitoring provided them with the raw material needed to codify employee expertise.

As AI increasingly enabled the codification of expertise across all sectors—from legal reasoning to medical diagnosis to financial analysis—labor markets may evolve toward this Hollywood model. The transformation of tacit human knowledge into explicit, tradable digital assets fundamentally reshaped work organization. Instead of permanent employment relationships, it was possible to view the rise of expertise markets where professionals provided their codified skills for specific applications while retaining rights over future uses—just as writers and actors had done in Hollywood for decades. Moreover, social media and online communication might shift towards personalized content generated by AI that had codified human likeness, life experiences and memories. These elements would serve as both training data and storytelling devices, feeding back as immersive content experiences, potentially with personalized ads embedded along the way. Beyond the workplace, this potential future raised the question of who should own the personal data that humans freely gave away by virtue of using online services and products. Do regular internet users constitute a ‘shadow workforce’ that contributes its codified expertise without compensation or credit to companies whose business model was to sell the fruits of that expertise back to the very people who created it?

Graham believed that in the future, regular individuals – and not just celebrities – would need to own and control their digital likeness or their digital expertise. “If you get a high-quality studio scan of your face and voice, that data can persist forever. The AI models will only improve, meaning that scan could be used to create an ultra-realistic version of you at any time in the future.”²⁸ To him, safeguarding personal data (especially the kind that could be used in AI training) was paramount. “When we create a version of someone which is so realistic, that you can’t tell the difference between that and the real world, we move into a space where we are extending reality – extending that person into a digital space.”²⁹

This raised crucial questions for Metaphysic and society at large: Could their ethical approach to AI content generation survive in industries without collective bargaining protections? As the technology to create hyper-realistic digital personas became commoditized, who would control the ‘rails of reality?’ Moreover, the technology’s ability to mix and match different aspects of performance—combining one performer’s physical movements with another’s voice, for instance—created entirely new categories of intellectual property challenges. How could ordinary workers protect their digital rights in an era where their every professional action could be captured, analyzed, and potentially recreated through AI?

Exhibit 1 Tom Cruise impersonator Miles Fisher (left) and the deepfake Tom Cruise created by Chris Ume (right)



Source: Chris Ume and Miles Fisher, Metaphysic.

Exhibit 2 Elvis Presley with the AGT Judges (2022)



Source: Extracted from America's Got Talent 2022, YouTube, January 2, 2023, <https://www.youtube.com/watch?v=nHDYpxYP6sk>, accessed February 2025.

Exhibit 3 Characters in Movie 'Here' at different ages



Source: Extracted from <https://metaphysic.ai/studios/here-movie/>, accessed January 2025.

Exhibit 4 Status of Digital Data Protection

In the digital world, companies recorded and analyzed a wide array of user actions to understand human behavior and monetize these insights. Sometimes monitoring was carried out by an employer conducting quality control, sometimes an operating system collected the small movements of an individual's mouse to create a unique digital identity. From social media posts to search history, and every digital interaction in between, human actions contributed to a growing digital footprint – at times without explicit consent. There were no U.S. federal laws protecting individuals from the commercialization of their digital trail and online behavior, other than a few exceptions (such as the data of minors).

This was in sharp contrast to Europe's GDPR legislation, which detailed when companies were allowed to process data and how that data should be managed. Additionally, there were no U.S. federal regulations that currently prohibit companies from using a person's image and likeness to train their AI models. The states of Tennessee and California passed laws protecting individuals from replication of their likeness; however, this still left their data vulnerable to becoming training data for generative AI models. For example, Meta AI and AI Creative Tools were trained on publicly available and licensed information, and more notably, on information provided to its platforms, such as photos and captions. This implied that each user of Facebook or other Meta services and platforms were at risk of having their data used for AI model training.

This lack of oversight benefitted technology companies; to train their AI models, they required extensive amounts of text, photos, videos, and other forms of data for the algorithm to recognize patterns. A single selfie from a social-media user would create a weak AI model as it lacked the information needed for the model to reproduce the variety of camera angles, lighting, and human phenotypes that existed in photos of people. Instead, companies aggregated data from millions of their users to enhance their models' capabilities. With better prediction, technology companies were able to profit by creating advertisements that better captured user attention. This cyclical process ensured that companies not only profited from the data that users provided – but they also leveraged AI to extract even more value from their users over time.

Source: Compiled by casewriters; Bloomberg Law, "Consumer Data Privacy Laws," <https://pro.bloomberglaw.com/insights/privacy/consumer-data-privacy-laws/#us-privacy-laws>, accessed November 2024; Munck Wilson Mandala, "AI Likeness Protections Legislative Update," Client Alerts, September 18, 2024, <https://www.munckwilson.com/news/ai-likeness-protections-legislative-update/>, accessed November 2024; Meta, "How Meta uses information for generative AI models and features," Privacy Centre, https://www.facebook.com/privacy/genai/?_rdr, accessed November 2024; European Council, "The general data protection regulation," <https://www.consilium.europa.eu/en/policies/data-protection-regulation/>, accessed January 2025.

Exhibit 5 Excerpts of Proposed Package to WGA

Category	Key Provisions
Guaranteed Minimum Employment	• Development Rooms: Writers guaranteed minimum of 10 weeks • Production Experience: Showrunners may select at least two writers for 20+ consecutive weeks • Required Employment: At least two mid-level writers for 20+ weeks
Artificial Intelligence Protections	• GAI-produced material will not be considered literary material • Writer's compensation, credit and separated rights not affected by GAI material ➢ GAI material will not be considered assigned material for compensation ➢ GAI material will not be considered source material for determining a writer's credit ➢ GAI material will not be basis for disqualifying a writer from eligibility of separated rights ➢ Companies must disclose if GAI created the material ➢ Writers cannot be required to use GAI to create literary material

Note: GAI (General Artificial Intelligence)

Source: Extracted from AMPTP Press Release, August 22, 2023, <https://labor.amptp.org/sites/labor.amptp.org/files/AMPTP%20Press%20Release%202023%2008%2022.pdf#overlay-context=node/19237>, accessed February 2025.

Exhibit 6 Upcoming Federal Legislation

In the wake of the Hollywood Guild negotiations, Congress took measures to forward the protection of worker data, although a comprehensive bill was yet to be passed by both chambers. In 2024, the NO FAKES Act was introduced as a means to address this gap in protections. The act was a bipartisan bill that would “protect the voice and visual likeness of all individuals” from generative AI creations. The main proposals included holding individuals or companies liable if they produced an unauthorized digital replica^d of an individual in a performance; holding platforms liable for hosting an unauthorized digital replica if the platform had actual knowledge of the fact that the replica was not authorized by the individual depicted; excluding certain digital replicas from coverage based on recognized First Amendment protections; and largely preempting State laws addressing digital replicas to create a workable national standard.

^d The definition of replica is “a newly-created, computer-generated, highly realistic electronic representation that is readily identifiable as the voice or visual likeness of an individual.” A digital replica refers to a sound recording, image, or audiovisual work where the individual either did not actually perform or appear, or where their performance or appearance has been significantly altered. However, standard practices like remixes, samples, mastering, or digital remastering of a performance, when authorized by the copyright holder, are not considered replicas.

Governor Gavin Newsom of California signed two bills to protect the likeness of individuals in audio and video production. AB 2602 “required contracts to specify the use of AI-generated digital replicas of a performer’s voice or likeness, and the performer must be professionally represented in negotiating the contract.” Meanwhile, AB 1836 “prohibits commercial use of digital replicas of deceased performers in films, TV shows, video games, audiobooks, sound recordings and more, without first obtaining the consent of those performers’ estates.”

Source: Compiled by casewriters; “Nurture Originals, Foster Art, and Keep Entertainment Safe (NO FAKES) Act,” https://www.coons.senate.gov/imo/media/doc/no_fakes_act_one-pager.pdf, accessed November 2024; 118th Congress, 2nd Session, “S. 4875: To protect intellectual property rights in the voice and visual likeness of individuals, and for other purposes,” <https://www.congress.gov/118/bills/s4875/BILLS-118s4875is.pdf>, accessed November 2024; Governor Gavin Newsom, “Governor Newsom signs bills to protect digital likeness of performers,” September 17, 2024, <https://www.gov.ca.gov/2024/09/17/governor-newsom-signs-bills-to-protect-digital-likeness-of-performers/>, accessed November 2024; Governor Gavin Newsom, “Governor Newsom signs bills to protect digital likeness of performers,” September 17, 2024, <https://www.gov.ca.gov/2024/09/17/governor-newsom-signs-bills-to-protect-digital-likeness-of-performers/>, accessed November 2024.

Exhibit 7 When Work Becomes Data

The evolution of digital monitoring had particular implications for workers. Employers were also increasingly gaining direct access to data on a broader range of employee tasks through tracking software on computers and digital work tools. Advancements in data storage and analytics had further intensified monitoring practices that existed for over a century, making algorithmic management of work processes more sophisticated and widespread. With improvements in digital tools available to employers, more employee data on-the-job was viable for collection and commercialization. An application of these enhanced capabilities was the use of employee work data to train machine learning and artificial intelligence models that could replicate an employee’s work. In certain cases, what appeared as consumer data was also worker data—for many creators, the data they produce served as the primary source of their livelihood. Every Tik-Tok video or Instagram ad published by micro-influencers and celebrities provided data to the respective companies governing the platforms.

Historically, a person’s output in the workplace was not so easily replicated—although certain tasks could be taught or systematized. For example, the artist Sol LeWitt, a trailblazer in conceptual art, created explicit instructions so that his artwork could be explicitly replicated for a fee. While LeWitt could offer instructions for past works, creating new works in his style was difficult. Replication was limited to the provision of clear instructions, which required LeWitt’s consent and input. With the creation of the computer and the codification of scientific management, work that previously required thorough training and knowledge was now more likely to be broken down into smaller, easier sub-tasks that could be assigned to one person. Taylorism made industry more efficient while making many workers more replaceable. But, tacit human skill and creativity could not be captured; instructions had to be explicitly coded.

There were two distinct cases of codified labor. In the first case, the data used as AI training data was a product of work that the employer and employee directly contract over. For example, a columnist for the *New York Times* was paid to produce articles for the newspaper. *The New York Times* could then use the past articles of its employees to produce new articles in the “style” of one of their columnists. The direct output of a columnist’s labor was therefore susceptible to codification, or to reproduction. In the second case, the data used to train an AI model of expertise was a byproduct of the employee’s labor and not directly contracted over. For example, employees of a venture capital firm who left a digital trail of presentations and recorded meetings, along with investment decisions and outcomes, were creating data to train an AI model of their expertise—but they were not directly paid for that

data product. In this instance, the VC employee may not be compensated for the very data that scaled their expertise, and potentially made their actions many times more valuable.

The application of new technologies in the collection and analysis of data from labor and its byproducts raised new questions surrounding the protections workers had (or did not have) to control the codification of their labor. Under copyright law in the United States, direct products produced by an employee were generally owned by their employer. This would imply that data produced by employees as a direct result of their job could be used to train AI models and potentially replicate the type of work conducted by employees within an organization. Currently, there were no federal laws restricting the employer's use of worker-generated data. In Illinois, a state with legislation limiting the use of biometric data, a lawsuit against Roosevelt University held that U.S. labor union workers cannot sue their employers over violations of state biometric data laws (and that instead they must settle such matters through union arbitration). This lawsuit showed that even in an instance of very sensitive data—biometric data—employees did not currently have rights to manage their data's use, even if the application fell outside of the scope of their job, or was used beyond the termination of their employment contract.

Source: Compiled by casewriters; Bernhardt, A., Kresge, L., and Suleiman, R., "The Data-Driven Workplace and the Case for Worker Technology Rights," *ILR Review*, 76(1), 3-29, 2023, <https://journals.sagepub.com/doi/10.1177/00197939221131558>, accessed November 2024; Belle Wong, "Does Your Employer Own Intellectual Property You Create," *Legal Zoom*, <https://www.legalzoom.com/articles/does-your-employer-own-intellectual-property-you-create>, accessed November 2024; GreenbergTraurig, "BIPA Update: Illinois Limits Liability and Clarifies Electronic Consent for Biometric Data Collection," *GT Alert*, August 14, 2024, <https://www.gtlaw.com/en/insights/2024/8/bipa-update-illinois-limits-liability-and-clarifies-electronic-consent-for-biometric-data-collection#:~:text=The%20Illinois%20General%20Assembly%20enacted,of%20hand%20or%20facial%20geometry>, accessed November 2024; U.S. News, "Union Workers Can't Sue Under Illinois Biometric Law, Court Rules," March 23, 2023, <https://www.usnews.com/news/top-news/articles/2023-03-23/union-workers-cant-sue-under-illinois-biometric-law-court-rules>, accessed November 2024.

Exhibit 8 Data Value and Attribution

Economists and policy-makers concerned with fairness and efficiency questioned whether the right incentives were in place to generate the most valuable data, and whether the efficient incentives would also compensate people for their proportional contribution to the AI models of expertise. Perhaps efficient incentives would even change the behavior of employees toward actions that further improved AI models, like experimenting more often to improve the quality of the training data.

In certain cases, figuring out individual contributions toward an AI model could be straightforward. Metaphysic used the voice and video of past performances by individuals to create the deepfake of their expertise, there was a relatively clear mapping between the data generated by the individual's past productions and the model rendering of the individual's expertise. There were challenges to verifying exactly what footage (potentially licensed from different studios) mattered most in the generation of the deepfake, but typically these could be resolved by examining the weights the model placed on distinct input data.

The music industry highlighted how difficult it was to trace the sources of data used in AI-generated works in the case of data recombination. EMI, a technology developed by professor of music at UC Santa Cruz, David Cope, used past compositions to rearrange pieces of the compositions into a new song while ensuring the new piece did not replicate any original work exactly - and, much the way artists build on prior work, the new composition was itself considered original. When first used to publish works inspired by Bach, the music industry was alarmed, with many critics claiming the music

produced by EMI was devoid of feeling and humanity. Realizing that what made composers relevant was their mortality, Cope discontinued EMI.

Its successor AI model, named Emily Howell, trained on EMI's output, recombined many pieces and produced her own original pieces in a style that could not be traced easily to any one artist. Unlike EMI, Emily had the added advantage of being able to modify works of music according to verbal input/suggestions. Cope highlighted the strength of using AI to deduce an implicit style or skill: "I couldn't figure out what my style was, so I suddenly started to say 'well, I don't know what my style is, so I should figure out what style is, period [with AI].'" EMI and Emily Howell were eventually able to codify Cope's style and reproduce new music in a recognizable style, before incorporating it and recombining it with others' styles. While technologies like Emily Howell once sparked controversy, AI often enhanced melodies, chord progressions, and lyrics, making data traceability more challenging and blurring lines around authenticity, ownership, and creativity.

In other cases, the contribution of an individual's data to the value of a model could be vanishingly small. In general, the more data collected by artificial intelligence models, the more effective they were at their intended use. However, once a large data corpus had been used to train a model, the marginal contribution of the next data on offer could be "de minimis," or so small as to be negligible. In other words, the collective data was valuable, but the marginal data was not.

Source: Compiled by casewriters; Bernard Marr, "EMI: When AIs Become Creative And Compose Music," Bernard Marr and Co., <https://bernardmarr.com/emi-when-ais-become-creative-and-compose-music/>, accessed November 2024; Chris Garcia, "Algorithmic Music - David Cope and EMI," *Computer History*, Blog Post, <https://computerhistory.org/blog/algorithmic-music-david-cope-and-emi/>, accessed November 2024.

Exhibit 9 New Compensation Models

Innovative approaches to data protection and compensation were emerging. One policy under consideration was the use of micropayments at the level of the data-piece; where the data could be a social media post, or a google search, or an address. In total, these micro payments could amount to meaningful financial transfers. The revenue generated by consumer data to Facebook and Google alone amounted to over \$600 per person living in the U.S., according to rough estimates. If the data of consumers and workers were to be monetized under user agreements and labor contracts, Americans could receive hundreds of dollars in payments across the distinct platforms profiting from their data. The perceived technological hurdle in protecting individual data is watermarking the data.

Toward this end, some had turned to blockchain technology. According to Susan McGregor, an Associate Research Scholar at Columbia University's Data Science Institute, "blockchain can allow artists to create a cryptographic log of their work," which allows them to prove ownership. This allows artists to take legal action against intellectual property violations. Artists were also taking advantage of new technologies like Glaze, a system designed by researchers at the University of Chicago to change the appearance of artworks to generative AI models (but not humans), essentially hiding the true appearance of the artwork to these models. In combination, these technologies helped track and protect the work of artists from exploitation, and when extended beyond the art realm, had the potential to modernize data protections into the era of generative AI.

Source: Compiled by casewriters; Ben Woford, "What's your data really worth?" *Proton*, Blog Post, February 8, 2024, <https://proton.me/blog/what-is-your-data-worth#:~:text=Depending%20where%20you%20live%2C%20information,analysis%20of%20their%20regulatory%20filings>, accessed November 2024; Anuradha Varanasi, "AI Art is Here to Stay - How Blockchain Can Help Creators Gain Control Over Their Work," Columbia University, Data Science Institute, March 28, 2023, <https://datascience.columbia.edu/news/2023/ai-art-is-here-to-stay-how-blockchain-can-help-creators-gain-control->

[over-their-work/](#), accessed November 2024; Glaze, "What is Glaze?" <https://glaze.cs.uchicago.edu/what-is-glaze.html>, accessed November 2024.

Endnotes

¹ Tom Graham, "Course: 3 Technologies That Will Change The World," Class Appearance at the Harvard Business School, February 21, 2023.

² SAG AFTRA Website, "About SAG-AFTRA," <https://www.sagaftra.org/spanish-language-television/about-sag-aftra#:~:text=SAG%2DAFTRA%20is%20a%20union,work%20in%20entertainment%20and%20media>, accessed November 2024.

³ Craig Silverman, "How to Spot A Deepfake Like The Barack Obama-Jordan Peele Video," *BuzzFeed*, April 17, 2018, <https://www.buzzfeed.com/craigsilverman/obama-jordan-peelee-deepfake-video-debunk-buzzfeed>, accessed January 2025.

⁴ Resemble.AI, "Justin Bieber AI-Generated Deepfake Song referencing Sean 'Diddy' Combs," <https://www.resemble.ai/justin-bieber-ai-generated-deepfake-song-referencing-sean-diddy-combs/>, accessed November 2024.

⁵ Tom Graham, "Course: 3 Technologies That Will Change The World," Class Appearance at the Harvard Business School, February 21, 2023.

⁶ James Vincent, "Tom Cruise deepfake creator says public shouldn't be worried about 'one-click fakes'," *The Verge*, March 5, 2021, <https://www.theverge.com/2021/3/5/22314980/tom-cruise-deepfake-tiktok-videos-ai-impersonator-chris-ume-miles-fisher>, accessed January 2025.

⁷ James Vincent, "Tom Cruise deepfake creator says public shouldn't be worried about 'one-click fakes'," *The Verge*, March 5, 2021, <https://www.theverge.com/2021/3/5/22314980/tom-cruise-deepfake-tiktok-videos-ai-impersonator-chris-ume-miles-fisher>, accessed January 2025.

⁸ KindredCast by LionTree, "The Future of Hollywood, Deepfakes & Your Rights in the AI Era with Metaphysic CEO Tom Graham," Podcast, January 16, 2024, <https://www.youtube.com/watch?v=XIBHVq4L-Tg&t=8sm>, accessed January 2025.

⁹ Dean Takahashi, "Metaphysic, AI startup behind Tom Cruise deepfakes, raises \$7.5M," *Venture Beat*, January 25, 2022, <https://venturebeat.com/games/metaphysic-ai-startup-behind-tom-cruise-deepfakes-raises-7-5m/>, accessed November 2024.

¹⁰ Beni Issembert, "Metaphysic's America's Got Talent experience, with Chris Ume and Tom Graham," *Metaphysic Blog*, June 6, 2022, <https://blog.metaphysic.ai/metaphysic-americas-got-talent/>, accessed January 2025.

¹¹ Beni Issembert, "Metaphysic's America's Got Talent experience, with Chris Ume and Tom Graham," *Metaphysic Blog*, June 6, 2022, <https://blog.metaphysic.ai/metaphysic-americas-got-talent/>, accessed January 2025.

¹² Tom Graham, "Course: 3 Technologies That Will Change The World," Class Appearance at the Harvard Business School, February 21, 2023.

¹³ Devin Gordon, "What if A.I. Is Actually Good for Hollywood?" *The New York Times*, November 1, 2024, <https://www.nytimes.com/2024/11/01/magazine/ai-hollywood-movies-cgi.html>, accessed January 2025.

¹⁴ Devin Gordon, "What if A.I. Is Actually Good for Hollywood?" *The New York Times*, November 1, 2024, <https://www.nytimes.com/2024/11/01/magazine/ai-hollywood-movies-cgi.html>, accessed January 2025.

¹⁵ Devin Gordon, "What if A.I. Is Actually Good for Hollywood?" *The New York Times*, November 1, 2024, <https://www.nytimes.com/2024/11/01/magazine/ai-hollywood-movies-cgi.html>, accessed January 2025.

¹⁶ Carolyn Giardina, "Tom Hanks, Robin Wright to Be De-Aged in Robert Zemeckis' New Movie Using Metaphysic AI Tool," *The Hollywood Reporter*, January 21, 2023, <https://www.hollywoodreporter.com/movies/movie-news/metaphysic-ai-tom-hanks-robin-wright-deaged-robert-zemeckis-caa-1235313318/>, accessed November 2024.

¹⁷ Devin Gordon, "What if A.I. Is Actually Good for Hollywood?" *The New York Times*, November 1, 2024, <https://www.nytimes.com/2024/11/01/magazine/ai-hollywood-movies-cgi.html>, accessed January 2025.

¹⁸ Devin Gordon, "What if A.I. Is Actually Good for Hollywood?" *The New York Times*, November 1, 2024, <https://www.nytimes.com/2024/11/01/magazine/ai-hollywood-movies-cgi.html>, accessed January 2025.

¹⁹ Rebecca Aizin, "Tom Hanks and Robin Wright to Be Digitally De-Aged in Upcoming Film from 'Forrest Gump' Director," *People*, Jan 31, 2023, <https://people.com/movies/tom-hanks-and-robin-wright-to-be-digitally-de-aged-in-upcoming-film->

from-forrest-gump-director/#:~:text=Hanks%2C%20and%20Wright%2C,over%20a%20number%20of%20years., accessed November 2024.

²⁰ KindredCast by LionTree, “The Future of Hollywood, Deepfakes & Your Rights in the AI Era with Metaphysic CEO Tom Graham,” Podcast, January 16, 2024, <https://www.youtube.com/watch?v=XIBHVq4L-Tg&t=8sm>, accessed January 2025.

²¹ KindredCast by LionTree, “The Future of Hollywood, Deepfakes & Your Rights in the AI Era with Metaphysic CEO Tom Graham,” Podcast, January 16, 2024, <https://www.youtube.com/watch?v=XIBHVq4L-Tg&t=8sm>, accessed January 2025.

²² Carl Franzen, “U.S. Copyright Office says AI generated content can be copyrighted – if a human contributes to or edits it,” *VentureBeat*, January 29, 2025, <https://venturebeat.com/ai/u-s-copyright-office-says-ai-generated-content-can-be-copyrighted-if-a-human-contributes-to-or-edits-it/>, accessed February 2025.

²³ KindredCast by LionTree, “The Future of Hollywood, Deepfakes & Your Rights in the AI Era with Metaphysic CEO Tom Graham,” Podcast, January 16, 2024, <https://www.youtube.com/watch?v=XIBHVq4L-Tg&t=8sm>, accessed January 2025.

²⁴ KindredCast by LionTree, “The Future of Hollywood, Deepfakes & Your Rights in the AI Era with Metaphysic CEO Tom Graham,” Podcast, January 16, 2024, <https://www.youtube.com/watch?v=XIBHVq4L-Tg&t=8sm>, accessed January 2025.

²⁵ Writers Guild of America West, “Know Your Right: Artificial Intelligence,” 2024, <https://www.wga.org/contracts/know-your-rights/artificial-intelligence>, accessed November 2024.

²⁶ SAG AFTRA Website, https://www.sagaftra.org/sites/default/files/sa_documents/TV-Theatrical_23_Summary_Agreement_Final.pdf, accessed November 2024.

²⁷ KindredCast by LionTree, “The Future of Hollywood, Deepfakes & Your Rights in the AI Era with Metaphysic CEO Tom Graham,” Podcast, January 16, 2024, <https://www.youtube.com/watch?v=XIBHVq4L-Tg&t=8sm>, accessed January 2025.

²⁸ KindredCast by LionTree, “The Future of Hollywood, Deepfakes & Your Rights in the AI Era with Metaphysic CEO Tom Graham,” Podcast, January 16, 2024, <https://www.youtube.com/watch?v=XIBHVq4L-Tg&t=8sm>, accessed January 2025.

²⁹ Tom Graham, “Course: 3 Technologies That Will Change The World,” Class Appearance at the Harvard Business School, February 21, 2023.